retrieve_eez

output →

data/eez/World_EEZ_v12_20231025_LR/eez_v12_lowres.gpkg

retrieve_osm_boundaries

output →

data/osm-boundaries/json/{country}_adm1.json

retrieve_jrc_ardeco

output →

data/jrc-ardeco/ARDECO-SNPTD.2021.table.csv
data/jrc-ardeco/ARDECO-SUVGDP.2021.table.csv

retrieve_nuts_2021_shapes

↪ input

.snakemake/storage/http/gisco-services.ec.europa.eu/distribution/v2/nuts/download/ref-nuts-2021-01m.geojson.zip

output →

data/nuts/NUTS_RG_01M_2021_4326_LEVL_0.geojson
data/nuts/NUTS_RG_01M_2021_4326_LEVL_1.geojson
data/nuts/NUTS_RG_01M_2021_4326_LEVL_2.geojson
data/nuts/NUTS_RG_01M_2021_4326_LEVL_3.geojson

build_osm_boundaries

↪ input

data/eez/World_EEZ_v12_20231025_LR/eez_v12_lowres.gpkg
data/osm-boundaries/json/{country}_adm1.json

output →

data/osm-boundaries/build/{country}_adm1.geojson

retrieve_databundle

output →

data/bundle/GDP_per_capita_PPP_1990_2015_v2.nc
data/bundle/corine/g250_clc06_V18_5.tif
data/bundle/eea/UNFCCC_v23.csv
data/bundle/emobility/KF2___count
data/bundle/emobility/PKW___count
data/bundle/gebco/GEBCO_2014_2D.nc
data/bundle/h2_salt_caverns_GWh_per_sqkm.geojson
data/bundle/je-e-21.03.02.xls
data/bundle/nama_10r_3popgdp.tsv.gz
data/bundle/natura/natura.tiff
data/bundle/ppp_2019_1km_Aggregated.tif

retrieve_osm_prebuilt

↪ input

.snakemake/storage/http/zenodo.org/records/14144752/files/buses.csv
.snakemake/storage/http/zenodo.org/records/14144752/files/converters.csv
.snakemake/storage/http/zenodo.org/records/14144752/files/lines.csv
.snakemake/storage/http/zenodo.org/records/14144752/files/links.csv
.snakemake/storage/http/zenodo.org/records/14144752/files/map.html
.snakemake/storage/http/zenodo.org/records/14144752/files/transformers.csv

output →

data/osm-prebuilt/0.6/buses.csv
data/osm-prebuilt/0.6/converters.csv
data/osm-prebuilt/0.6/lines.csv
data/osm-prebuilt/0.6/links.csv
data/osm-prebuilt/0.6/map.html
data/osm-prebuilt/0.6/transformers.csv

build_shapes

↪ input

data/bundle/GDP_per_capita_PPP_1990_2015_v2.nc
data/bundle/ppp_2019_1km_Aggregated.tif
data/eez/World_EEZ_v12_20231025_LR/eez_v12_lowres.gpkg
data/jrc-ardeco/ARDECO-SNPTD.2021.table.csv
data/jrc-ardeco/ARDECO-SUVGDP.2021.table.csv
data/nuts/NUTS_RG_01M_2021_4326_LEVL_3.geojson
data/osm-boundaries/build/BA_adm1.geojson
data/osm-boundaries/build/MD_adm1.geojson
data/osm-boundaries/build/UA_adm1.geojson
data/osm-boundaries/build/XK_adm1.geojson

output →

resources/country_shapes.geojson
resources/europe_shape.geojson
resources/nuts3_shapes.geojson
resources/offshore_shapes.geojson

base_network

↪ input

<function input_base_network at 0x7b3501f30fe0>
resources/country_shapes.geojson
resources/europe_shape.geojson
resources/nuts3_shapes.geojson
resources/offshore_shapes.geojson

output →

resources/admin_shapes.geojson
resources/networks/base.nc
resources/regions_offshore.geojson
resources/regions_onshore.geojson

build_transmission_projects

↪ input

<input function>
resources/europe_shape.geojson
resources/networks/base.nc
resources/offshore_shapes.geojson

output →

resources/transmission_projects/adjust_lines.csv
resources/transmission_projects/adjust_links.csv
resources/transmission_projects/new_buses.csv
resources/transmission_projects/new_lines.csv
resources/transmission_projects/new_links.csv

retrieve_electricity_demand

output →

data/electricity_demand_raw.csv

retrieve_synthetic_electricity_demand

↪ input

.snakemake/storage/http/zenodo.org/records/10820928/files/demand_hourly.csv

output →

data/load_synthetic_raw.csv

add_transmission_projects_and_dlr

↪ input

<input function>
<input function>
resources/networks/base.nc

output →

resources/networks/base_extended.nc

simplify_network

↪ input

resources/admin_shapes.geojson
resources/networks/base_extended.nc
resources/regions_offshore.geojson
resources/regions_onshore.geojson

output →

resources/busmap_base_s.csv
resources/networks/base_s.nc
resources/regions_offshore_base_s.geojson
resources/regions_onshore_base_s.geojson

build_electricity_demand

↪ input

<input function>
data/electricity_demand_raw.csv

output →

resources/electricity_demand.csv

build_electricity_demand_base

↪ input

resources/electricity_demand.csv
resources/networks/base_s.nc
resources/nuts3_shapes.geojson
resources/regions_onshore_base_s.geojson

output →

resources/electricity_demand_base_s.nc

cluster_network

↪ input

<function input_custom_busmap at 0x7b3501e4aca0>
<input function>
resources/admin_shapes.geojson
resources/electricity_demand_base_s.nc
resources/networks/base_s.nc
resources/regions_offshore_base_s.geojson
resources/regions_onshore_base_s.geojson

output →

resources/busmap_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
resources/linemap_base_s_{clusters,[0-9]+(m|c)?|all|adm}.csv
resources/networks/base_s_{clusters,[0-9]+(m|c)?|all|adm}.nc
resources/regions_offshore_base_s_{clusters,[0-9]+(m|c)?|all|adm}.geojson
resources/regions_onshore_base_s_{clusters,[0-9]+(m|c)?|all|adm}.geojson

build_austrian_network

output →

resources/networks/austria.nc

prepare_residual_inputs

↪ input

data/MA/loads/NIP2030/GrossWasserkraft.csv
data/MA/loads/NIP2030/Kleinwasserkraft.csv
data/MA/loads/NIP2030/Nachfrage.csv
data/MA/loads/NIP2030/PV.csv
data/MA/loads/NIP2030/Wind.csv
data/MA/loads/NIP2030/biomasse.csv

output →

residual/NIP2030/residual_inputs.parquet

merge_austrian_network

↪ input

resources/networks/austria.nc
resources/networks/base_s_{clusters}.nc

output →

resources/networks/base_s_{clusters,[0-9]+(m|c)?|all|adm}_merged.nc

map_residual_to_buses

↪ input

residual/NIP2030/residual_inputs.parquet
resources/networks/base_s_128_merged.nc

output →

residual/NIP2030/mapped_residual_inputs.parquet

build_powerplants

↪ input

data/custom_powerplants.csv
resources/networks/base_s_{clusters}.nc

output →

resources/powerplants_s_{clusters,[0-9]+(m|c)?|all|adm}.csv

retrieve_cost_data

output →

resources/costs_{year}.csv

add_electricity

↪ input

<function input_conventional at 0x7b3501e4976d>
<function input_profile_tech at 0x7b3501e4b920>
<input function>
<input function>
data/hydro_capacities.csv
data/unit_commitment.csv
resources/busmap_base_s_{clusters}.csv
resources/electricity_demand_base_s.nc
resources/networks/base_s_{clusters}.nc
resources/powerplants_s_{clusters}.csv
resources/regions_onshore_base_s_{clusters}.geojson

output →

resources/networks/base_s_{clusters,[0-9]+(m|c)?|all|adm}_elec.nc

inject_residual_loads

↪ input

residual/NIP2030/mapped_residual_inputs.parquet
resources/networks/base_s_128_elec.nc

output →

resources/networks/NIP2030/base_s_128_elec_residualload.nc

retrieve_ship_raster

↪ input

.snakemake/storage/http/zenodo.org/records/13757228/files/shipdensity_global.zip

output →

data/shipdensity_global.zip

retrieve_cutout

↪ input

.snakemake/storage/http/zenodo.org/records/14936211/files/{cutout}.nc

output →

cutouts/{cutout}.nc

build_ship_raster

↪ input

<input function>
data/shipdensity_global.zip

output →

resources/shipdensity_raster.tif

build_hydro_profile

↪ input

<input function>
data/eia_hydro_annual_capacity.csv
data/eia_hydro_annual_generation.csv
data/era5-annual-runoff-per-country.csv
resources/country_shapes.geojson

output →

resources/profile_hydro.nc

download_wdpa

↪ input

.snakemake/storage/http/d1gam3xoknrgr2.cloudfront.net/current/WDPA_Jun2025_Public_shp.zip

output →

data/WDPA.gpkg

download_wdpa_marine

↪ input

.snakemake/storage/http/d1gam3xoknrgr2.cloudfront.net/current/WDPA_WDOECM_Jun2025_Public_marine_shp.zip

output →

data/WDPA_WDOECM_marine.gpkg

download_copernicus_land_cover

↪ input

.snakemake/storage/http/zenodo.org/records/3939050/files/PROBAV_LC100_global_v3.0.1_2019-nrt_Discrete-Classification-map_EPSG-4326.tif

output →

data/Copernicus_LC100_global_v3.0.1_2019-nrt_Discrete-Classification-map_EPSG-4326.tif

determine_availability_matrix_MD_UA

↪ input

<input function>
<input function>
<input function>
<input function>
data/Copernicus_LC100_global_v3.0.1_2019-nrt_Discrete-Classification-map_EPSG-4326.tif
data/WDPA.gpkg
data/WDPA_WDOECM_marine.gpkg
resources/country_shapes.geojson
resources/offshore_shapes.geojson

output →

resources/availability_matrix_MD-UA_{clusters,[0-9]+(m|c)?|all|adm}_{technology}.nc

determine_availability_matrix

↪ input

<function <lambda> at 0x7b3501e4b240>
<function input_ua_md_availability_matrix at 0x7b3501f31760>
<input function>
<input function>
<input function>
<input function>
<input function>
data/bundle/corine/g250_clc06_V18_5.tif
resources/country_shapes.geojson
resources/offshore_shapes.geojson

output →

resources/availability_matrix_{clusters,[0-9]+(m|c)?|all|adm}_{technology}.nc

build_renewable_profiles

↪ input

<input function>
resources/availability_matrix_{clusters}_{technology}.nc
resources/offshore_shapes.geojson
resources/regions_onshore_base_s_{clusters}.geojson

output →

resources/profile_{clusters,[0-9]+(m|c)?|all|adm}_{technology,(?!hydro).*}.nc